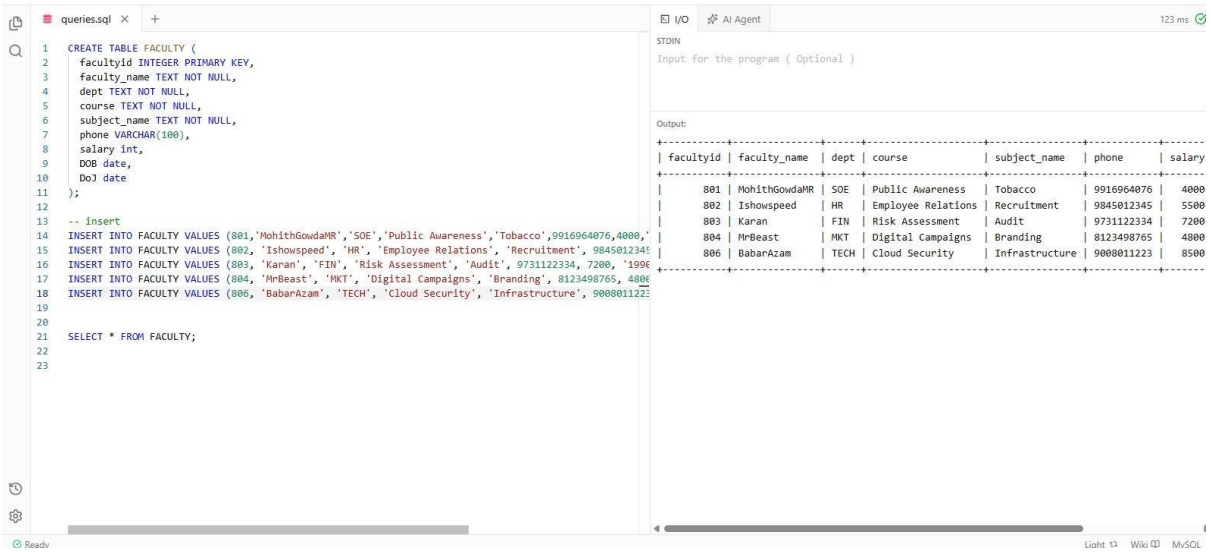
NAME:Yashas.S
USN:1RUA25BCA0120

DBMS

1.CREATING THE TABLE



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,'
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 984501234
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '199
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 480
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 900801122

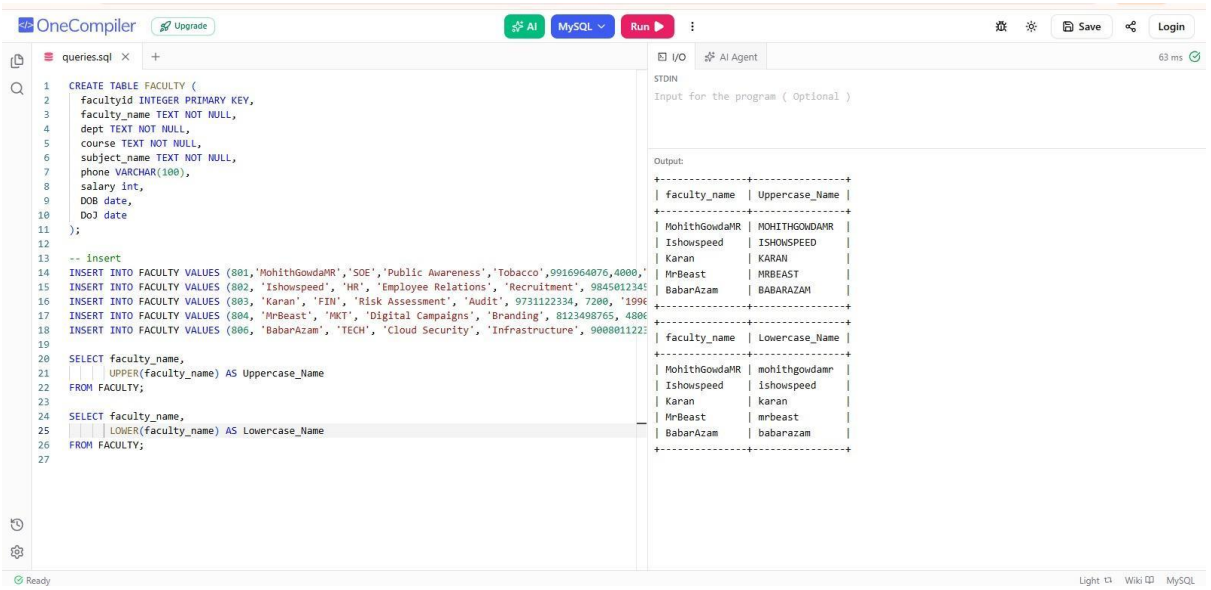

SELECT * FROM FACULTY;
```

```
| facultyid | faculty_name | dept | course              | subject_name   | phone      | salary
|       801 | MohithGowdaMR | SOE  | Public Awareness    | Tobacco        | 9916964076 |   4000
|       802 | Ishowspeed    | HR   | Employee Relations  | Recruitment    | 9845012345 |   5500
|       803 | Karan         | FIN  | Risk Assessment     | Audit          | 9731122334 |   7200
|       804 | MrBeast       | MKT  | Digital Campaigns   | Branding       | 8123498765 |   4800
|       806 | BabarAzam     | TECH | Cloud Security      | Infrastructure | 9008011223 |   8500
```

STRING FUNCTIONS

2.UPPER CASE

3.LOWER CASE



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,'
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 984501234
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '199
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 480
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 900801122

SELECT faculty_name,
       UPPER(faculty_name) AS Uppercase_Name
FROM FACULTY;

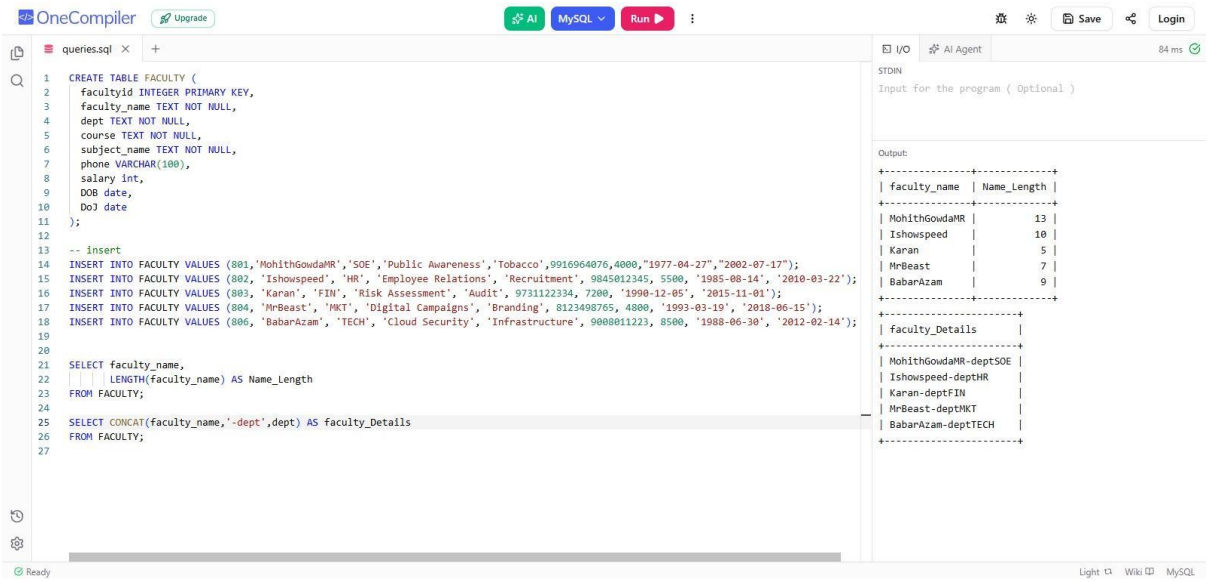
SELECT faculty_name,
       LOWER(faculty_name) AS Lowercase_Name
FROM FACULTY;
```

```
| faculty_name  | Uppercase_Name |
| MohithGowdaMR | MOHITHGOWDAMR  |
| Ishowspeed    | ISHOWSPEED     |
| Karan         | KARAN          |
| MrBeast       | MRBEAST        |
| BabarAzam     | BABARAZAM      |

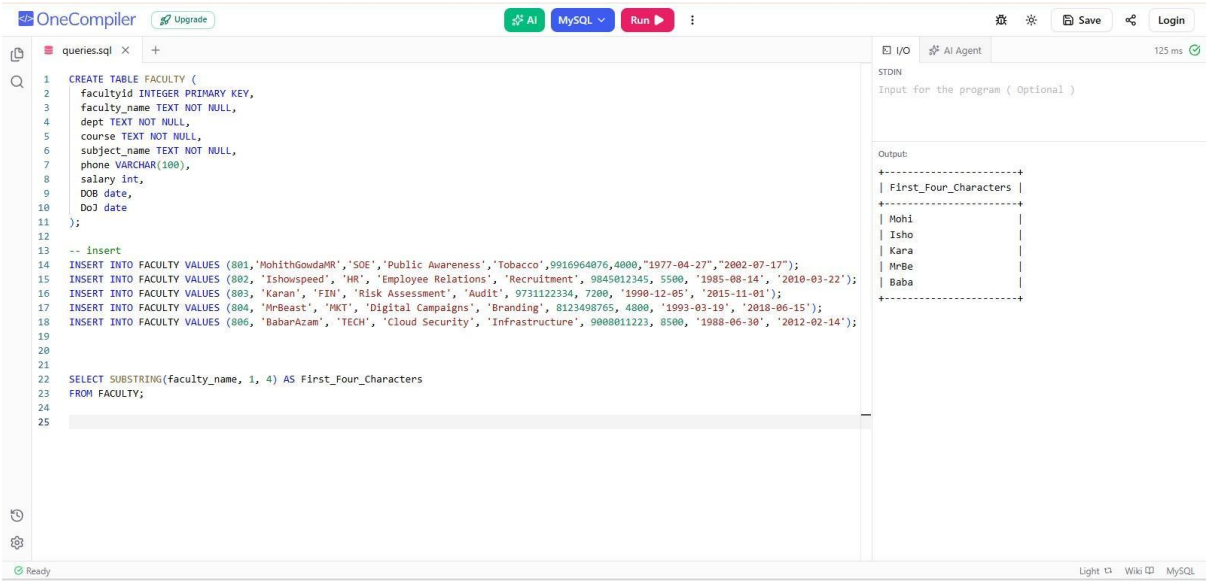
| faculty_name  | Lowercase_Name |
| MohithGowdaMR | mohithgowdamr  |
| Ishowspeed    | ishowspeed     |
| Karan         | karan          |
| MrBeast       | mrbeast        |
| BabarAzam     | babarazam      |
```

4.LENGTH()

5.CONCAT()

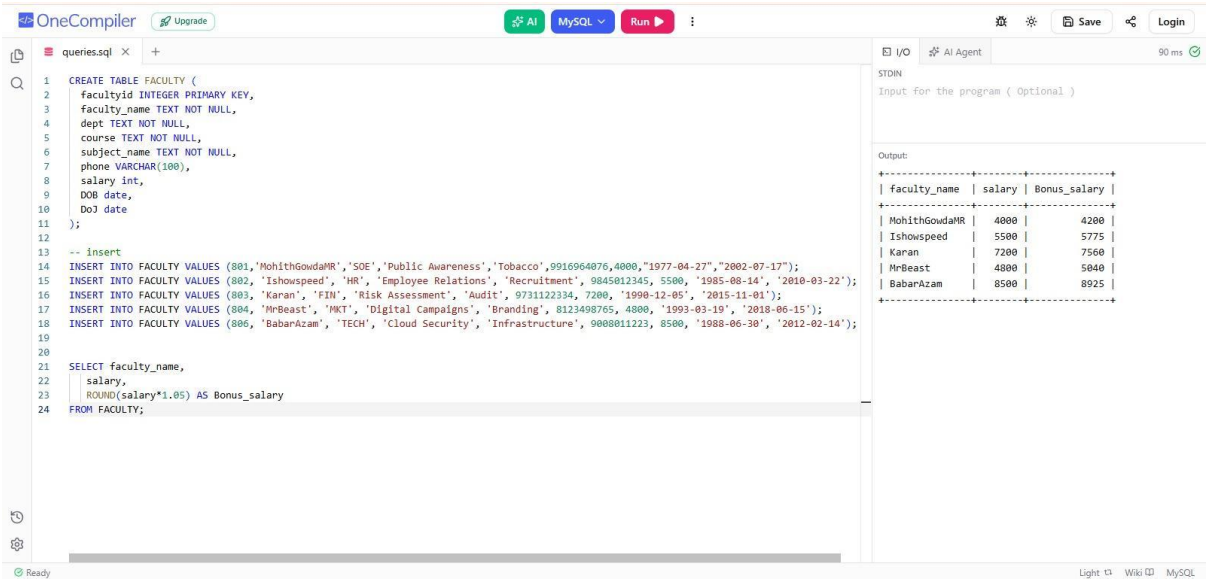


6.SUBSTRING()



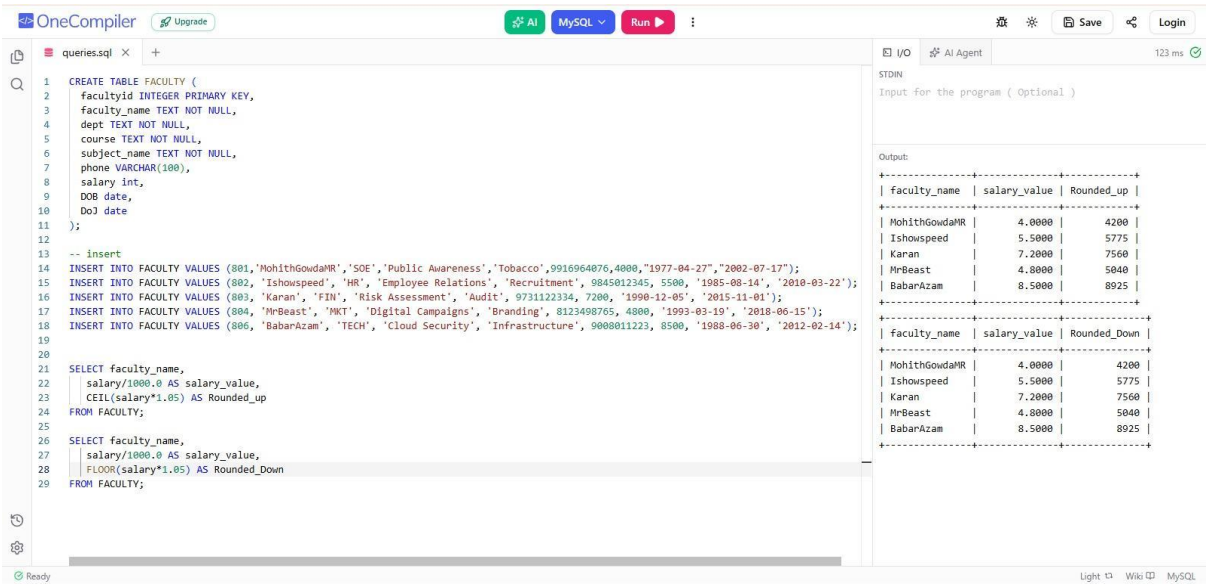
NUMERIC FUNCTIONS

7.ROUND()



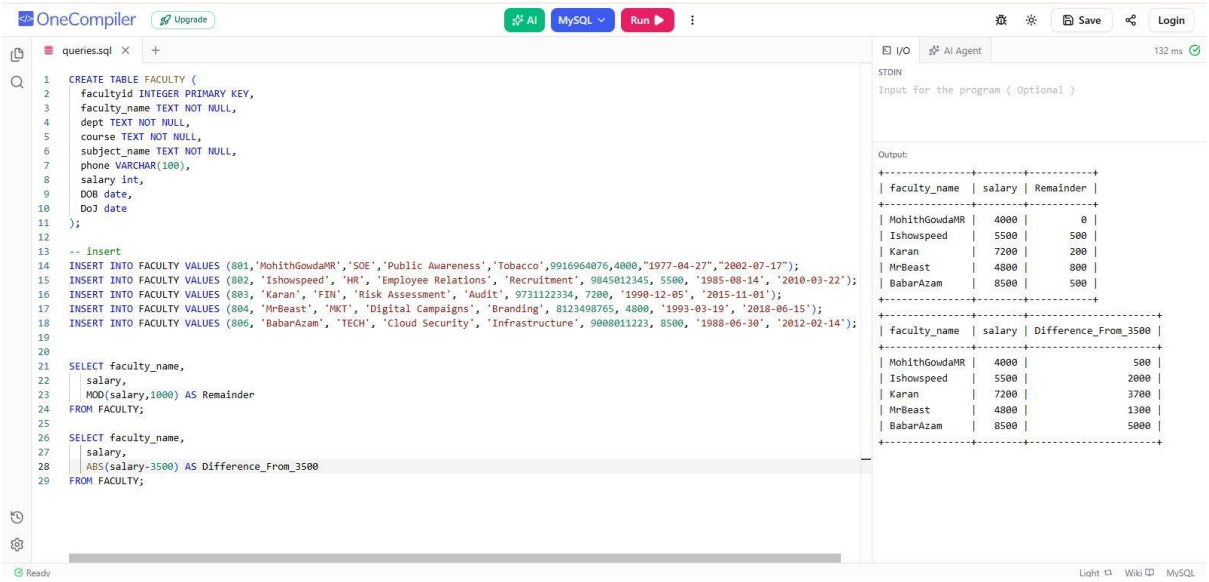
8.CEIL()

9.FLOOR()



10.MOD()

11.ABS()



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-27","2002-07-17");
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '1985-08-14', '2010-03-22');
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05', '2015-11-01');
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-03-19', '2018-06-15');
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '1988-06-30', '2012-02-14');


SELECT faculty_name,
    salary,
    MOD(salary,1000) AS Remainder
FROM FACULTY;

SELECT faculty_name,
    salary,
    ABS(salary-3500) AS Difference_From_3500
FROM FACULTY;
```

```
+--------------+--------+-----------+
| faculty_name | salary | Remainder |
+--------------+--------+-----------+
| MohithGowdaMR |   4000 |         0 |
| Ishowspeed    |   5500 |       500 |
| Karan         |   7200 |       200 |
| MrBeast       |   4800 |       800 |
| BabarAzam     |   8500 |       500 |
+--------------+--------+-----------+
+--------------+--------+----------------------+
| faculty_name | salary | Difference_From_3500 |
+--------------+--------+----------------------+
| MohithGowdaMR |   4000 |                  500 |
| Ishowspeed    |   5500 |                 2000 |
| Karan         |   7200 |                 3700 |
| MrBeast       |   4800 |                 1300 |
| BabarAzam     |   8500 |                 5000 |
+--------------+--------+----------------------+
```

DATE FUNCTIONS

12.EXTRACTING THE YEAR()

13.EXTRACTING THE MONTH()

14.EXTRACTING THE DAY()



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-27","2002-07-17");
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '1985-08-14', '2010-03-22');
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05', '2015-11-01');
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-03-19', '2018-06-15');
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '1988-06-30', '2012-02-14');

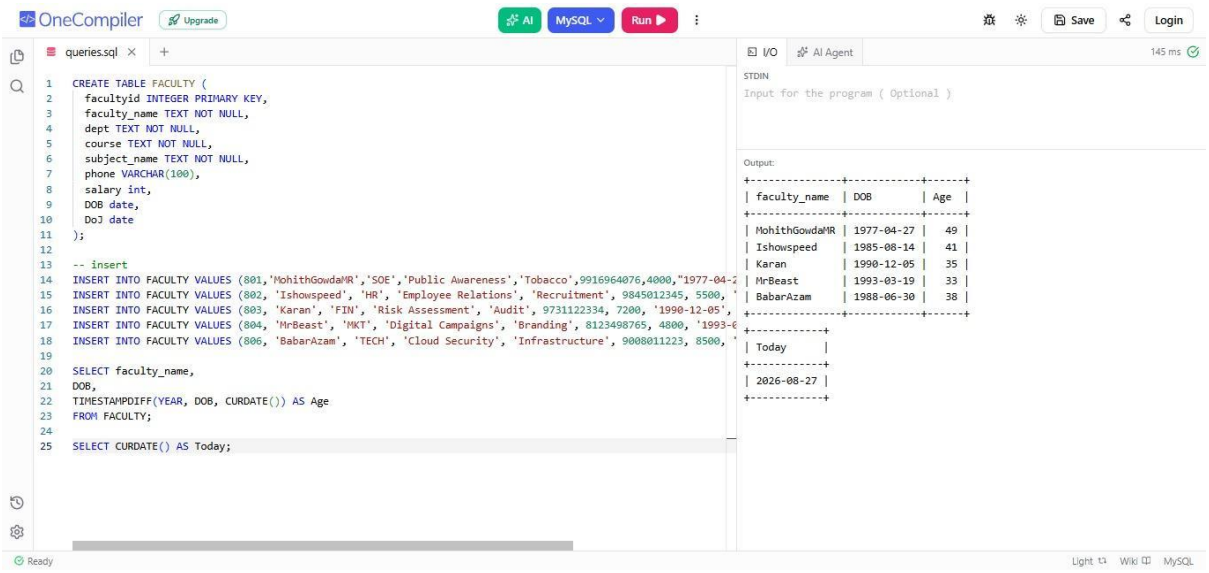
SELECT faculty_name,
    DOB,
    YEAR(DOB) AS Birth_year
FROM FACULTY;

SELECT faculty_name,
    DOB,
    MONTH(DOB) AS Birth_Month
FROM FACULTY;

SELECT faculty_name,
    DOB,
    DAY(DOB) AS Birth_Month
FROM FACULTY;
```

```
+--------------+------------+------------+
| faculty_name | DOB        | Birth_year |
+--------------+------------+------------+
| MohithGowdaMR | 1977-04-27 |       1977 |
| Ishowspeed    | 1985-08-14 |       1985 |
| Karan         | 1990-12-05 |       1990 |
| MrBeast       | 1993-03-19 |       1993 |
| BabarAzam     | 1988-06-30 |       1988 |
+--------------+------------+------------+
+--------------+------------+-------------+
| faculty_name | DOB        | Birth_Month |
+--------------+------------+-------------+
| MohithGowdaMR | 1977-04-27 |           4 |
| Ishowspeed    | 1985-08-14 |           8 |
| Karan         | 1990-12-05 |          12 |
| MrBeast       | 1993-03-19 |           3 |
| BabarAzam     | 1988-06-30 |           6 |
+--------------+------------+-------------+
+--------------+------------+-------------+
| faculty_name | DOB        | Birth_Month |
+--------------+------------+-------------+
| MohithGowdaMR | 1977-04-27 |          27 |
| Ishowspeed    | 1985-08-14 |          14 |
| Karan         | 1990-12-05 |           5 |
| MrBeast       | 1993-03-19 |          19 |
| BabarAzam     | 1988-06-30 |          30 |
+--------------+------------+-------------+
```

15.CALCULATING APPROXIMATE AGE()



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-2
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05',
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-0
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '

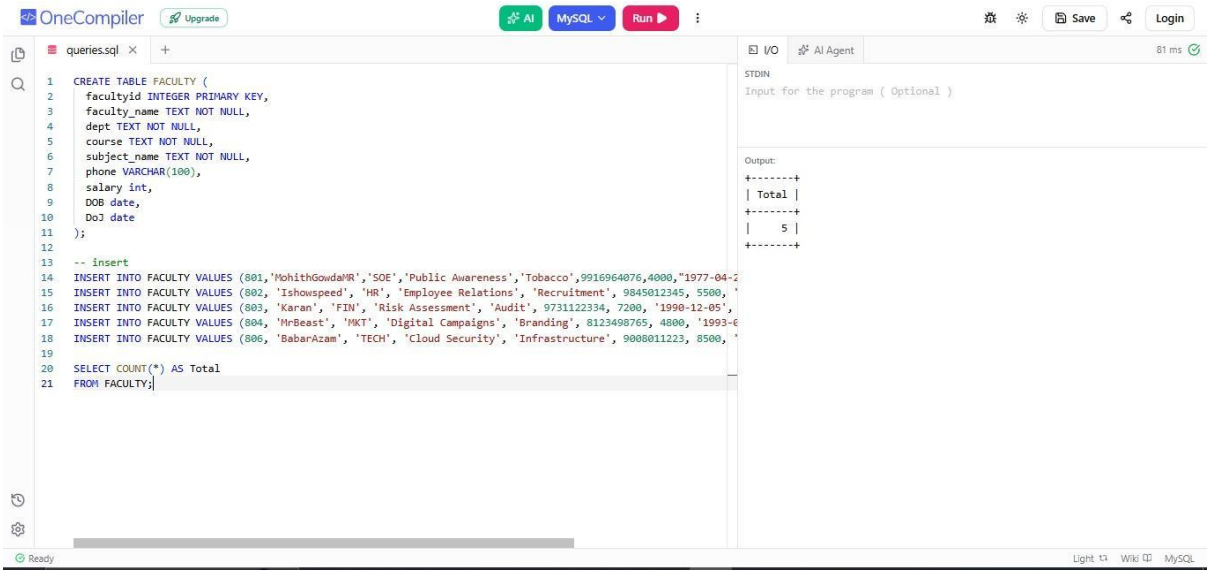
SELECT faculty_name,
DOB,
TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age
FROM FACULTY;

SELECT CURDATE() AS Today;
```

```
+--------------+------------+------+
| faculty_name | DOB        | Age  |
+--------------+------------+------+
| MohithGowdaMR | 1977-04-27 |   49 |
| Ishowspeed   | 1985-08-14 |   41 |
| Karan        | 1990-12-05 |   35 |
| MrBeast      | 1993-03-19 |   33 |
| BabarAzam    | 1988-06-30 |   38 |
+--------------+------------+------+
+------------+
| Today      |
+------------+
| 2026-08-27 |
+------------+
```

AGGREGATE FUNCTIONS

16.COUNT(*)



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-2
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05',
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-0
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '

SELECT COUNT(*) AS Total
FROM FACULTY;
```

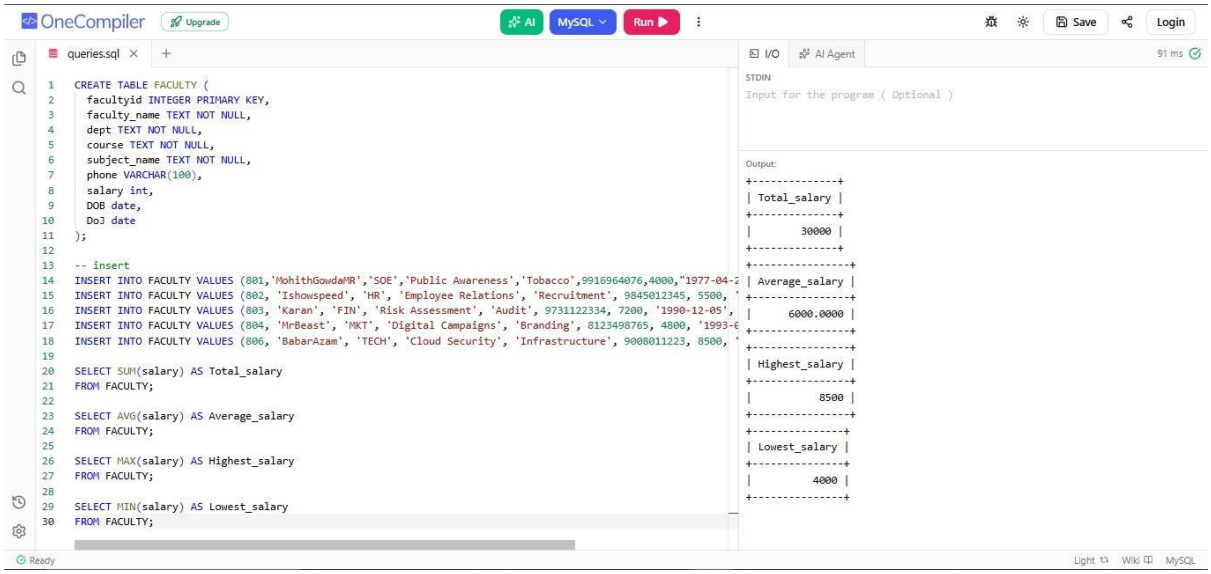
```
+-------+
| Total |
+-------+
|     5 |
+-------+
```

17.SUM()

18.AVG()

19.MIN()

20.MAX()



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-2
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05',
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-6
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '

SELECT SUM(salary) AS Total_salary
FROM FACULTY;

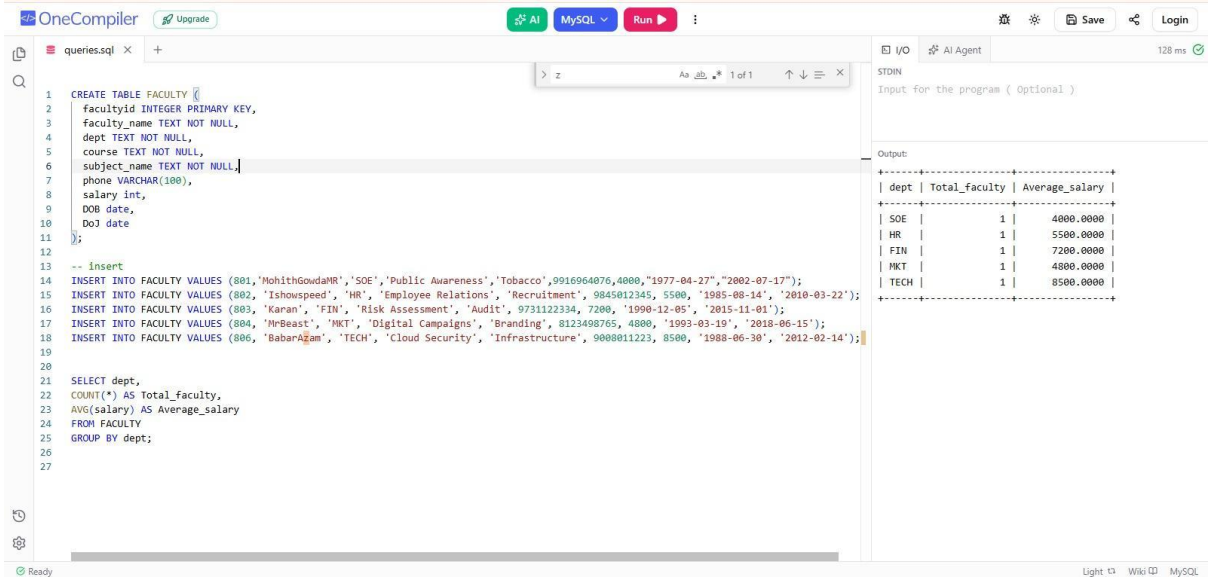
SELECT AVG(salary) AS Average_salary
FROM FACULTY;

SELECT MAX(salary) AS Highest_salary
FROM FACULTY;

SELECT MIN(salary) AS Lowest_salary
FROM FACULTY;
```

```
+--------------+
| Total_salary |
+--------------+
|        30000 |
+--------------+
+----------------+
| Average_salary |
+----------------+
|      6000.0000 |
+----------------+
+----------------+
| Highest_salary |
+----------------+
|           8500 |
+----------------+
+--------------+
| Lowest_salary |
+--------------+
|          4000 |
+--------------+
```

21.GROUP BY()



```
CREATE TABLE FACULTY (
  facultyid INTEGER PRIMARY KEY,
  faculty_name TEXT NOT NULL,
  dept TEXT NOT NULL,
  course TEXT NOT NULL,
  subject_name TEXT NOT NULL,
  phone VARCHAR(100),
  salary int,
  DOB date,
  DoJ date
);

-- insert
INSERT INTO FACULTY VALUES (801,'MohithGowdaMR','SOE','Public Awareness','Tobacco',9916964076,4000,"1977-04-27","2002-07-17");
INSERT INTO FACULTY VALUES (802, 'Ishowspeed', 'HR', 'Employee Relations', 'Recruitment', 9845012345, 5500, '1985-08-14', '2010-03-22');
INSERT INTO FACULTY VALUES (803, 'Karan', 'FIN', 'Risk Assessment', 'Audit', 9731122334, 7200, '1990-12-05', '2015-11-01');
INSERT INTO FACULTY VALUES (804, 'MrBeast', 'MKT', 'Digital Campaigns', 'Branding', 8123498765, 4800, '1993-03-19', '2018-06-15');
INSERT INTO FACULTY VALUES (806, 'BabarAzam', 'TECH', 'Cloud Security', 'Infrastructure', 9008011223, 8500, '1988-06-30', '2012-02-14');


SELECT dept,
COUNT(*) AS Total_faculty,
AVG(salary) AS Average_salary
FROM FACULTY
GROUP BY dept;
```

```
+------+---------------+----------------+
| dept | Total_faculty | Average_salary |
+------+---------------+----------------+
| SOE  |             1 |      4000.0000 |
| HR   |             1 |      5500.0000 |
| FIN  |             1 |      7200.0000 |
| MKT  |             1 |      4800.0000 |
| TECH |             1 |      8500.0000 |
+------+---------------+----------------+
```